

Physio Companion

Mobile Physiotherapy Assistant Application, Providing Real-Time Feedback and Performance Tracking for Knee Rehabilitation



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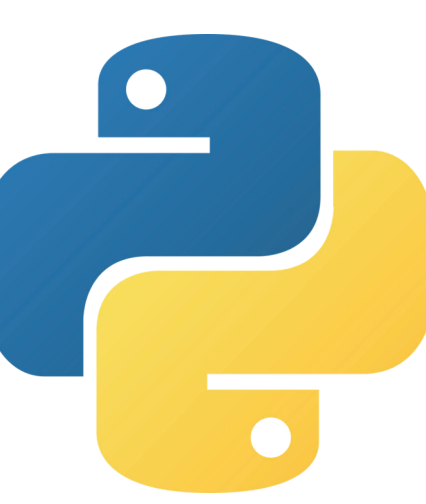
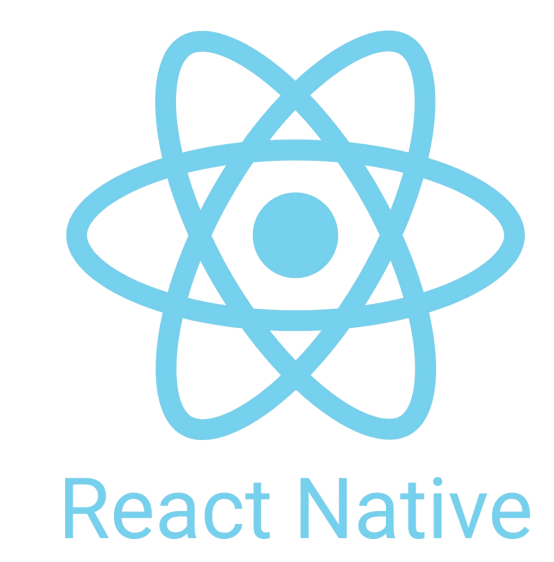
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Background

- Over 2.4 billion people worldwide will require physical rehabilitation at some point in their lives [Cieza et al., 2021]
- Many patients struggle performing their prescribed physiotherapy exercises correctly once they return home, often leading to poor form and less effective recovery

Overview

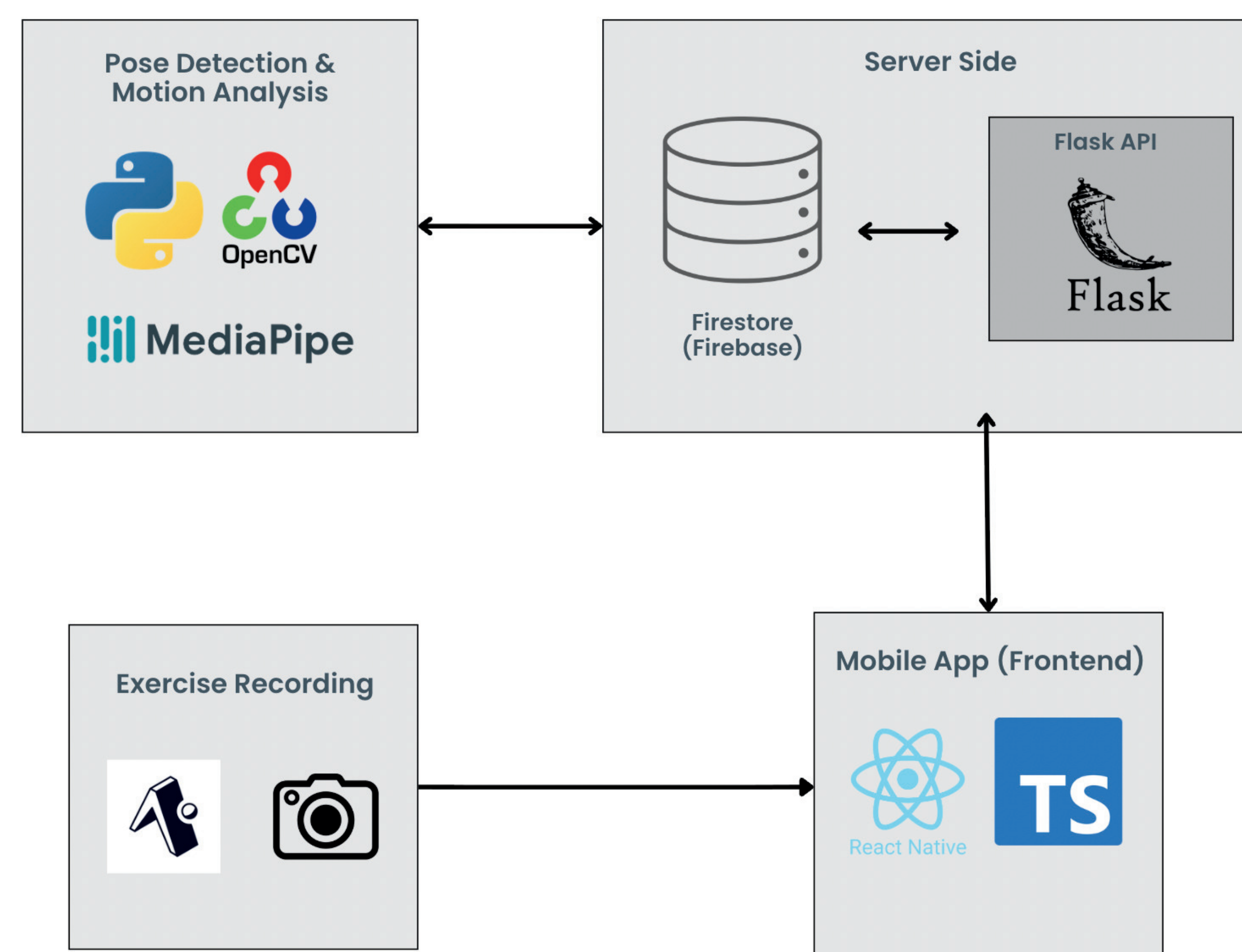
- Physio Companion is a mobile physiotherapy assistant application that helps patients complete their physiotherapy exercises on their own
- Uses computer vision to track key body movement to provide real-time feedback during exercises
- Tracks key performance metrics (i.e. repetitions and ROM), to allow users the ability to monitor their progress over time

Goals

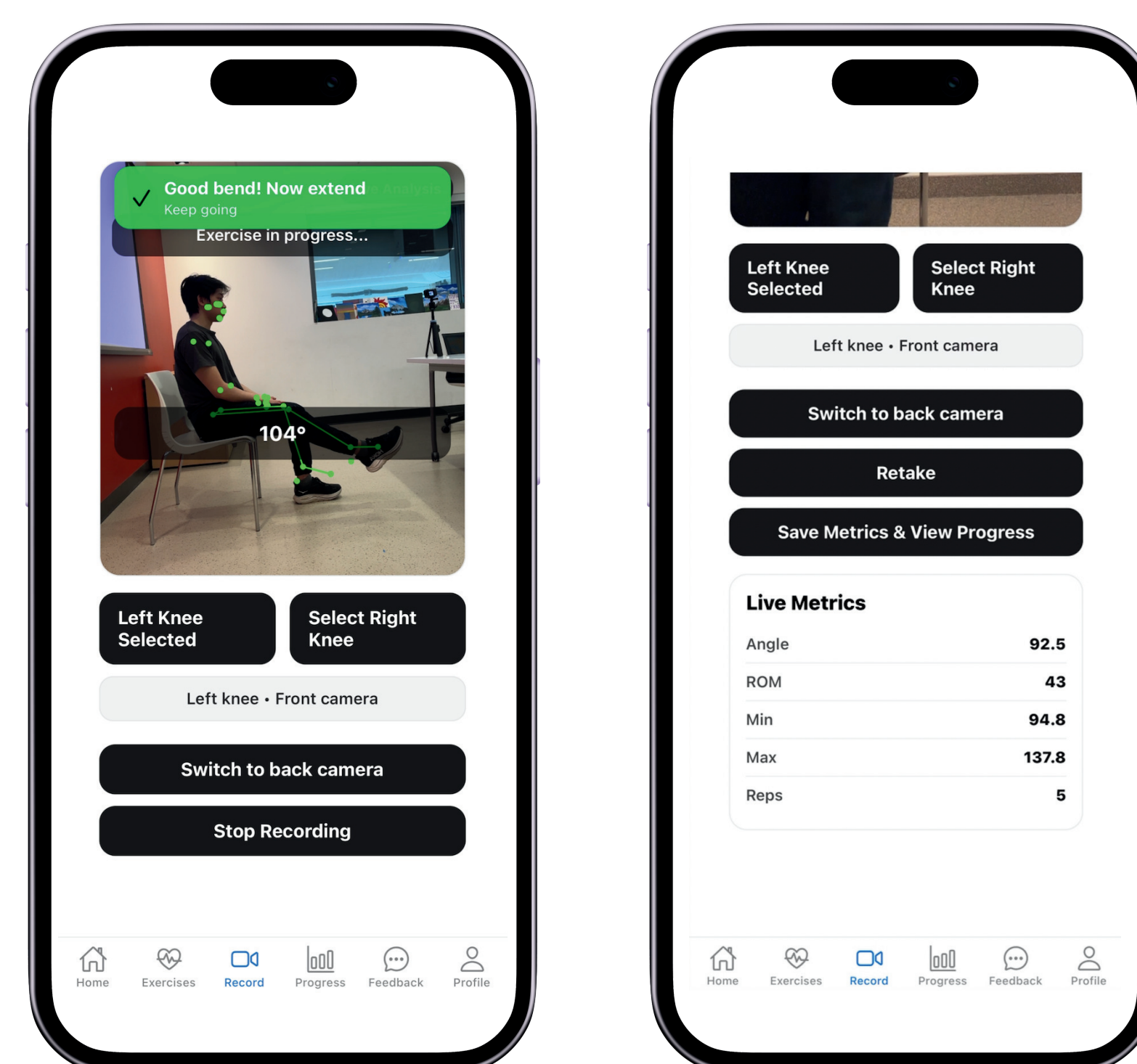
- At-Home Physiotherapy Support:** Enables patients to complete prescribed exercises on their own without constant supervision
- Provides Real-Time Exercise Feedback:** Provides patients with live feedback to allow them to better understand how well they are performing each movement
- Builds Patient Confidence & Consistency:** Feedback and progress tracking encourages patients to stay motivated and stay on track with their exercises
- User-Friendly Interface:** Designed for easy navigation and is accessible to all users



Design Overview



Pose Detection & Motion Analysis

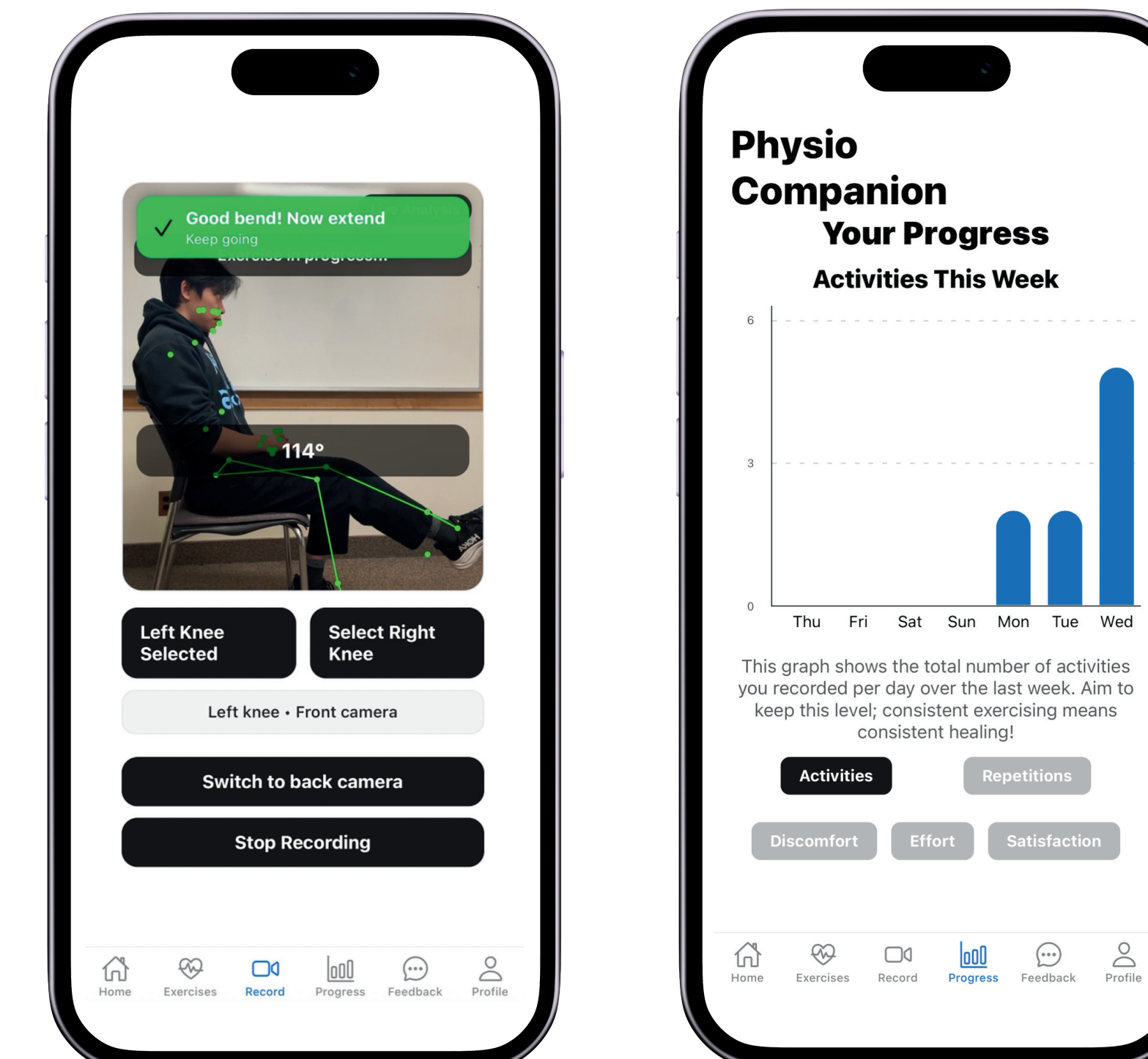


- MediaPipe identifies **key body landmarks (hip, knee, ankle)** to compute the **knee angle** from each video frame
- OpenCV** processes frames to **track movement** and **extract relevant data**
- Knee joint angles are calculated to analyze extracted data and compute **performance metrics (repetition count, ROM, etc.)**

Key Features



Exercise Feedback & Results



- Displays **real-time visual feedback** (i.e. Good Bend! Now extend) to correct form and motivate the user
- Provides **audio cues to guide users** (i.e. when to start the exercise or the current repetition count)
- Displays performance metrics through **graphical views** to help users **track progress over time**

Usability Results

Usability testing was conducted on **5 participants**: 2 physiotherapists and 3 other individuals that represented a subset of the diverse general population. The conclusions drawn from the study were:

- Audio feedback provided by the application would **enhance user experience**
- Additional metrics derived from the recording analysis would **provide physiotherapists with more accurate insights**
- The **UI was intuitive** and **easy to navigate**
- Overall perceived usability of the system was **high**

Acknowledgements

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- Special thanks to all individuals who participated in our usability testing

Future Work

- Improved Pose Detection Accuracy:** Enhance the precision of movement tracking to provide more accurate exercise feedback
- Expanded Exercise Library:** Support additional exercises beyond single lower body part, covering full-body rehabilitation exercises
- Gamification & Engagement:** Introduce additional features such as achievement badges, or reminders to improve user motivation and adherence